

Adaptive Heuristic Portfolio TSP Report

Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8_full_solver_evolution.
- Best combined transfer gap: phase8_full_solver_evolution.
- Lowest selector regret: phase8_full_solver_evolution.
- Pareto-efficient conditions: phase8_full_solver_evolution, phase8_random_portfolio, phase8_supervised_ml_selector.

Run Metadata

- run_name: run_20260521_000102_s
- started_at_local: 2026-05-21 00:01:02
- finished_at_local: 2026-05-21 00:33:43
- duration_hhmm: 00:33
- duration_seconds: 1960.041
- seed_offset: 18000
- replicate_label: s
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime-Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_best_single_fixed_heuristic	best_fixed	none	0.07226	0.0	1481.613257	0.07226	0.007284	0.03122	0.0	0.0	False
phase8_random_portfolio	random_portfolio	none	0.141534	0.069274	295.251714	0.141534	0.065695	0.09363	0.0	0.0	True
phase8_oracle_selector	oracle_selector	none	0.07226	0.0	1506.255043	0.073462	8.9e-05	0.026678	0.0	0.0	False

Condition	Mode	Replay	Final TSPLI B Gap	Selector Regret	Runtime (ms)	Runtime -Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_supervised_ml_selector	supervised_selector	none	0.07226	0.0	1475.330914	0.07226	0.007284	0.03122	0.0	0.0	True
phase8_llm_static_selector	static_selector	none	0.21578	0.14352	439.288357	0.21578	0.133367	0.16373	0.0	0.62	False
phase8_llm_evolved_adaptive_controller	adaptive_controller	none	0.084733	0.012473	4081.4391	0.233416	0.015088	0.04074	0.146159	0.62	False
phase8_llm_evolved_controller_diversity_failure_replay	adaptive_controller	diversity_failure	0.084733	0.012473	4085.646214	0.233657	0.014551	0.04047	0.494316	0.62	False
phase8_full_solver_evolution	full_solver	none	0.072172	-8.8e-05	461.409171	0.072172	0.0	0.02659	0.785065	0.76	True

Condition Notes

phase8_best_single_fixed_heuristic

- Execution mode best_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1481.613257 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.
- Fixed heuristic choice: farthest_insertion_two_opt.

phase8_random_portfolio

- Execution mode random_portfolio with replay none.
- Final held-out gap 0.141534, selector regret 0.069274, runtime-adjusted gap 0.141534.
- Family transfer 0.065695 and combined transfer 0.093636.
- Runtime 295.251714 ms, runtime inflation -0.800723, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_oracle_selector

- Execution mode oracle_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.073462.
- Family transfer 8.9e-05 and combined transfer 0.026678.

- Runtime 1506.255043 ms, runtime inflation 0.016632, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_supervised_ml_selector

- Execution mode supervised_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1475.330914 ms, runtime inflation -0.00424, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_llm_static_selector

- Execution mode static_selector with replay none.
- Final held-out gap 0.21578, selector regret 0.14352, runtime-adjusted gap 0.21578.
- Family transfer 0.133367 and combined transfer 0.16373.
- Runtime 439.288357 ms, runtime inflation -0.703507, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cheapest_insertion_two_opt:edge_preserving_restart:candidate_pruned_two_opt:stag4:cand-2:restart1.
- Controller rule summary: {"acceptance_bias": 0.006, "candidate_limit_offset": -2, "clustered_heuristic": "cheapest_insertion_two_opt", "corridor_heuristic": "cluster_first_local_search", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-conditioned frozen-portfolio selector tuned for general EUC_2D structure with bottleneck and grid/corridor awareness.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "farthest_insertion_two_opt", "low_improvement_threshold": 0.07, "name": "tsp_hh_oracle_compass_v1", "nearest_neighbor_trap_heuristic": "annealed_multi_start", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "candidate_pruned_two_opt", "stagnation_threshold": 4, "temperature_scale": 1.15, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"}

phase8_llm_evolved_adaptive_controller

- Execution mode adaptive_controller with replay none.
- Final held-out gap 0.084733, selector regret 0.012473, runtime-adjusted gap 0.233416.
- Family transfer 0.015088 and combined transfer 0.040747.
- Runtime 4081.4391 ms, runtime inflation 1.754726, mean novelty 0.146159, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:limited_three_opt:annealed_multi_start:stag4:cand-2:restart2.
- Controller rule summary: {"acceptance_bias": 0.008, "candidate_limit_offset": -2, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "edge_preserving_restart", "default_heuristic": "farthest_insertion_two_opt", "description": "Interpretable instance-adaptive scheduler over a frozen heuristic portfolio with conservative stagnation and perturbation control.", "failed_perturbation_threshold": 3, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.12, "name": "deterministic_instance_adaptive_tsp_hh_v3", "nearest_neighbor_trap_heuristic": "farthest_insertion_two_opt", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 2, "stagnation_switch_heuristic": "annealed_multi_start", "stagnation_threshold": 4, "temperature_scale": 1.35, "time_budget_trigger": 0.6,

"two_cluster_bottleneck_heuristic": "limited_three_opt"}

phase8_llm_evolved_controller_diversity_failure_replay

- Execution mode adaptive_controller with replay diversity_failure.
- Final held-out gap 0.084733, selector regret 0.012473, runtime-adjusted gap 0.233657.
- Family transfer 0.014551 and combined transfer 0.040407.
- Runtime 4085.646214 ms, runtime inflation 1.757566, mean novelty 0.494316, and complexity 0.62.
- Pareto efficient: False.
- Controller signature:
annealed_multi_start:cluster_first_local_search:limited_three_opt:edge_preserving_restart:stag3:cand-3:restart2.
- Controller rule summary: {"acceptance_bias": 0.012, "candidate_limit_offset": -3, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "cluster_first_local_search", "default_heuristic": "farthest_insertion_two_opt", "description": "Interpretable instance-adaptive frozen-portfolio controller with earlier diversification and conservative specialization under diversity-failure replay.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.12, "name": "deterministic_instance_adaptive_tsp_hh_v3", "nearest_neighbor_trap_heuristic": "farthest_insertion_two_opt", "random_like_heuristic": "annealed_multi_start", "restart_offset": 2, "stagnation_switch_heuristic": "edge_preserving_restart", "stagnation_threshold": 3, "temperature_scale": 1.35, "time_budget_trigger": 0.45, "two_cluster_bottleneck_heuristic": "limited_three_opt"}

phase8_full_solver_evolution

- Execution mode full_solver with replay none.
- Final held-out gap 0.072172, selector regret -8.8e-05, runtime-adjusted gap 0.072172.
- Family transfer 0.0 and combined transfer 0.02659.
- Runtime 461.409171 ms, runtime inflation -0.688577, mean novelty 0.785065, and complexity 0.76.
- Pareto efficient: True.

Judge Appendix

Conservative Interpretation of TSP Adaptive Heuristic Portfolio Results

Metric	phase8_best_single_fixed_heuristic	phase8_oracle_selector	phase8_supervised_ml_selector	phase8_llm_evolved_adaptive_controller	phase8_full_solver_evolution
Held-out TSPLIB gap	0.07226	0.07226	0.07226	0.08473	**0.07217**
Selector regret vs oracle	0.0	0.0	0.0	0.01247	**-8.8e-05**
Runtime-adjusted gap	0.07226	0.07346	0.07226	0.23342	**0.07217**
Runtime inflation	0.0	0.01663	-0.00424	1.75	**-0.69**
Final transfer gap	0.03122	0.02668	0.03122	0.04075	**0.02659**
Pareto efficient	No	No	Yes	No	Yes

Metric	phase8_best_single_fixed_heuristic	phase8_oracle_selector	phase8_supervised_ml_selector	phase8_llm_evolved_adaptive_controller	phase8_full_solver_evolution
Mean complexity	0.0	0.0	0.0	0.62	0.76

Key Points

1. **Primary Endpoint (Held-out TSPLIB gap):**

- The `*phase8_full_solver_evolution*` condition achieves the lowest held-out gap (0.07217), marginally better than the oracle selector and best single heuristic (both 0.07226).
- This suggests it closely approximates oracle performance on TSPLIB instances, the main evaluation criterion.

2. **Selector Regret vs Oracle:**

- `*phase8_full_solver_evolution*` also shows negligible negative regret ($-8.8e-05$), effectively matching oracle selector performance in selection quality.
- Adaptive controllers have small positive regret, indicating slightly worse than oracle but still relatively close.

3. **Runtime-Adjusted Gap & Runtime Inflation:**

- `*phase8_full_solver_evolution*` has runtime-adjusted gap comparable to best single heuristic and supervised ML selector but with **substantial runtime reduction (-69%)** relative to oracle selector, indicating efficient performance without runtime inflation.
- Adaptive controllers show notably higher runtime inflation ($\sim 1.75x$) and significantly worse runtime-adjusted gaps (~ 0.23), which limits their deployability despite their adaptivity.

4. **Pareto Efficiency:**

- `*phase8_full_solver_evolution*`, the supervised ML selector, and the random portfolio lie on the Pareto frontier balancing solution quality and runtime metrics.
- Static or adaptive selectors that approach oracle accuracy without excessive runtime inflation are prioritized; here, `*phase8_full_solver_evolution*` best meets this.

5. **Cross-Family Transfer (Transfer Gap):**

- `*phase8_full_solver_evolution*` exhibits the lowest transfer gap (0.02659), indicating better generalization across TSP families compared to others, including adaptive controllers and best fixed heuristic.

6. **Adaptation & Controller Interpretability:**

- Adaptive controllers show modest adaptation efficiency but suffer in runtime cost and final TSPLIB gap.
- `*phase8_full_solver_evolution*`, although more complex (complexity = 0.76), provides interpretable instance-adaptive hyper-heuristic control with strong performance and low runtime inflation.
- No novel algorithms discovered; improvements stem from effective hyper-heuristic control over known heuristics.

Summary

- ****The *phase8_full_solver_evolution* condition is the most balanced and effective approach**, achieving the best held-out TSPLIB gap with minimal selector regret vs oracle and significant runtime efficiency gains.**
- It maintains Pareto efficiency, strong cross-family generalization, and interpretable, instance-adaptive portfolio control without excessive runtime inflation.
- Adaptive controllers provide interesting adaptation but at the cost of higher runtimes and worse overall TSPLIB gaps, limiting their practical advantage.
- The static supervised ML selector is competitive but does not surpass the *full_solver_evolution* in transfer or runtime efficiency.
- The oracle selector remains a theoretical upper bound; *phase8_full_solver_evolution* closely approximates it and can be considered the leading deployable strategy.

Recommendation

Prioritize the *phase8_full_solver_evolution* approach as the best instance-adaptive hyper-heuristic control for TSP given its balance of solution quality, runtime efficiency, and cross-family transfer, without claiming novel heuristic discovery. Further refinement should focus on maintaining interpretability while controlling runtime inflation in adaptive controllers.